Ocean Freight Rate & Shipment Tracker

Problem Statement

TEAM SIZE	DURATION	TECH STACK	DELIVERABLES
Individual	2 days	Any of your choice	4 items

Background

Importers and exporters keep shipping rates and shipment details in spreadsheets and emails. This makes it hard to find the cheapest carrier, track shipments and catch billing mistakes. Build a simple web application that fixes this.

Task

Build a web application with the following four modules.

1

Rate Search

- Load carrier rates from a CSV file provided by the organizers.
- The user picks origin port, destination port and container type (20ft or 40ft).
- The app lists the matching carriers, sorted by price, with transit time shown.
- The cheapest option is highlighted.

2

Shipment Tracker

- The user can add a shipment (ID, carrier, origin, destination, departure date, expected arrival).
- A dashboard lists all shipments with a status: *In Transit*, *Delivered* or *Delayed*.
- A shipment is marked *Delayed* when today's date is past its expected arrival and it is not delivered.

3

Invoice Checker

- The user enters or uploads an invoice amount for a shipment.
- The app compares it to the contracted rate and shows *Correct* or *Overcharged by INR/USD X*.

4

Email-Based Rate & Shipment Updates

- Some rate changes and shipment status updates arrive as plain email text, not CSV. Parse a set of sample emails (.txt files) to extract the fields, instead of manual entry.
- Rate-update emails update the rate table; shipment-update emails update the tracker status.
- A sample inbox of 10 to 15 emails is provided, including a few irrelevant ones that should be skipped, not crash the app.

Basic Requirements

- Login page with hashed passwords (two roles: Admin and User).
- Any tech stack (for example HTML/CSS/JS with Python, Node or Java, and MySQL, SQLite or MongoDB).
- Sample data provided (carrier rates CSV and a few sample shipments).

Sample data provided

`carrier_rates.csv` carrier, origin_port, destination_port, container_type, rate_usd, transit_days

`sample_shipments.csv` shipment_id, carrier, origin_port, destination_port, container_type, departure_date, expected_arrival, delivered_date, invoice_amount_usd

`sample_emails.txt` 10 to 15 sample email texts (rate updates, shipment status updates and a few irrelevant emails to filter out)

Deliverables

Working demo
Local or hosted.

Source code
With a short README.

3-slide presentation
Approach and tech stack.

3-minute demo
Live walkthrough of your app.

Evaluation Rubric

Criteria	Weight
Working modules (rate search, tracker, invoice checker)	30%
Email-based update parsing	20%
Code quality and database design	15%
User interface and ease of use	10%
Login and basic security	10%
Presentation and teamwork	15%
Total	**100%**

Optional Bonus (not required)

- Chart of shipments by status.
- Filter or search on the shipment list.
- Export results to Excel.